

GRADE 8 · READING INFORMATIONAL TEXT · RI.8.5

The Magnetic Map: How Sea Turtles Navigate Home

Analyze in detail how a specific paragraph in an informational text is structured, identifying the role each sentence plays in developing and refining a key concept.

Lesson Map · Passage · Guided · Partner · Independent Practice · Small Group · Answer Key

What You Will Learn Today

Learning Goals

Understand RI.8.5: Analyze in detail how a specific paragraph in an informational text is...

Read an original passage closely

Support every answer with text evidence

Work through guided, partner, and independent practice

How the Lesson Runs

Warm-up and passage

Guided practice together

Partner discussion

Independent + small-group work

Exit ticket and answer key

Unpacking the Standard

Every well-built paragraph has sentences that do different jobs: a topic sentence that introduces the key concept, supporting sentences that provide evidence or explanation, a sentence that refines or complicates the idea, and sometimes a transition sentence that connects to what follows. At grade 8, identify not just what a paragraph says, but what job each individual sentence is doing.

Modeled Example

In a paragraph, the first sentence states, "Coral reefs face a growing threat from ocean acidification." What job is this sentence most likely doing?

It is most likely the topic sentence, introducing the paragraph's key concept (ocean acidification as a threat to coral reefs) that the rest of the paragraph will develop.

If a later sentence in that same paragraph says, "However, some coral species have shown surprising resilience in early trials," what job is that sentence doing relative to the topic sentence?

It is refining or complicating the topic sentence's claim, adding nuance by showing the threat is not uniform across all coral species, rather than simply repeating or supporting the original claim.

How Sea Turtles Navigate Home

1. Every year, loggerhead sea turtles hatch on beaches along the southeastern United States, swim thousands of miles across the open Atlantic, and eventually return to nest on the very same stretch of coastline where they were born, sometimes after more than a decade at sea. For years, scientists had a name for this ability but not a full explanation: natal homing.
2. Researchers now believe the answer lies in a sense most humans do not have: magnetoreception, the ability to detect Earth's magnetic field. Every location on Earth has a slightly different magnetic signature, based on the angle and strength of the planet's magnetic field at that point. A sea turtle hatchling appears to imprint on the specific magnetic signature of its home beach within its first few weeks of life, the way some animals imprint visually on a parent.
3. That imprinted signature becomes something like an internal map. As the turtle travels thousands of miles over years, it can sense whether it is north, south, east, or west of its home signature, adjusting its swimming direction accordingly, the same way a hiker might use a compass without needing to see the destination directly.

How Sea Turtles Navigate Home

4. A 2015 study tested this idea directly. Researchers exposed hatchling loggerheads to magnetic fields matching two different real locations along the Atlantic coast, then observed which direction the hatchlings tried to swim. The hatchlings consistently oriented themselves as if trying to move toward the coastline associated with their actual birth location, even though they had never experienced that magnetic field difference through any other sense.

5. The theory is not without complications. Magnetic signatures drift slightly over time due to natural shifts in Earth's magnetic field, meaning a turtle's internal map, formed once in infancy, could theoretically become slightly mismatched with its home beach's current signature decades later. Some researchers propose this drift may explain rare cases of turtles nesting on beaches near, but not exactly at, their birth location.

6. Despite these open questions, magnetoreception remains the leading explanation for one of the animal kingdom's most precise long-distance navigation feats, turning an ordinary beach into something closer to a magnetic address memorized before a turtle takes its first swim.

Warm-up · Before You Read

1. What explicit fact does paragraph 1 state about the distance loggerhead turtles travel before returning to nest?

2. What is the topic sentence of paragraph 2, and what key concept does it introduce?

Guided Practice

3. What job does the second sentence of paragraph 2 ("Every location on Earth has a slightly different magnetic signature...") perform relative to the topic sentence?

4. What job does the third sentence of paragraph 2 ("A sea turtle hatchling appears to imprint...") perform?

Guided Practice

5. What is the topic sentence of paragraph 3, and how does it build directly on paragraph 2's final idea?

6. What job does the comparison to a hiker's compass, at the end of paragraph 3, perform in the paragraph's structure?

Partner Discussion

7. What is the topic sentence of paragraph 4, and what evidence does the rest of the paragraph provide to support it?

8. Which sentence in paragraph 4 states the study's specific result, and how does it connect back to the magnetoreception concept from paragraph 2?

Partner Discussion

9. What job does the first sentence of paragraph 5 ("The theory is not without complications.") perform in the paragraph's structure?

Independent Practice

10. What specific complication does paragraph 5 introduce, and which sentence explains its possible real-world effect?

11. How does the first sentence of paragraph 6 ("Despite these open questions...") function as a transition, and what two ideas does it connect?

Independent Practice

12. What job does the article's final phrase, comparing a beach to "a magnetic address memorized before a turtle takes its first swim," perform?

13. Choose paragraph 2 and explain, sentence by sentence, how each sentence builds toward a fuller understanding of magnetoreception.

Independent Practice

14. Why might the article place the 2015 study's evidence (paragraph 4) after the explanation of magnetoreception (paragraphs 2–3) rather than before it?

15. Explain how paragraph 5's structure (introducing a complication, then explaining its possible effect) differs from paragraph 2's structure (introducing a concept, then explaining it directly). What is the purpose of each structure?

Small-Group Work

16. If paragraph 5 (the complications) were removed entirely, how would that change the overall structure and credibility of the article?

17. What explicit fact does paragraph 1 state about how scientists previously understood natal homing?

Small-Group Work

18. How does the first sentence of paragraph 1 set up the concept that paragraph 2's topic sentence will answer?

19. What is the topic sentence of paragraph 6, and how does it function differently from the topic sentences of paragraphs 2 through 5?

Exit Ticket

20. Explain how the structure of paragraphs 4 and 5 together mirrors the structure of scientific reasoning: presenting evidence, then testing its limits.

Answer Key (1 of 4)

1. They swim thousands of miles across the open Atlantic, sometimes returning after more than a decade at sea.
2. The topic sentence is "Researchers now believe the answer lies in a sense most humans do not have: magnetoreception," which introduces magnetoreception as the key concept the rest of the paragraph will explain.
3. It is a supporting sentence that provides background explanation needed to understand the topic sentence's claim, defining what a "magnetic signature" is before the paragraph can explain how turtles use it.
4. It is a supporting sentence that applies the general concept of magnetic signatures specifically to sea turtles, explaining how and when the imprinting process happens.
5. "That imprinted signature becomes something like an internal map" is the topic sentence; it builds directly on paragraph 2's idea of imprinting by explaining what the imprinted signature is then used for.

Answer Key (2 of 4)

6. It is a clarifying/supporting sentence that uses a familiar analogy to make the abstract concept of magnetic navigation easier to understand, refining the reader's understanding of the internal map concept.
7. "A 2015 study tested this idea directly" is the topic sentence; the rest of the paragraph provides the study's method (exposing hatchlings to different magnetic fields) and its result (hatchlings oriented toward their birth location) as supporting evidence.
8. "The hatchlings consistently oriented themselves as if trying to move toward the coastline associated with their actual birth location" states the result, directly confirming the magnetoreception and imprinting concepts introduced earlier in the article.
9. It is a refining/complicating sentence that signals a shift from supporting evidence to acknowledging limitations, preparing the reader for a more nuanced view of the theory.
10. The complication is that magnetic signatures drift over time; the sentence "Some researchers propose this drift may explain rare cases of turtles nesting on beaches near, but not exactly at, their birth location" explains its possible real-world effect.

Answer Key (3 of 4)

11. It connects the complications raised in paragraph 5 back to the overall strength of the magnetoreception theory, acknowledging uncertainty while reaffirming the theory's status as the leading explanation.

12. It is a concluding sentence that restates the article's key concept in a memorable, figurative way, giving the reader a final, vivid image that ties together imprinting, magnetic signatures, and navigation.

13. Sentence 1 introduces magnetoreception as the key concept; sentence 2 explains the background fact (magnetic signatures vary by location) needed to understand it; sentence 3 explains specifically how sea turtles use this ability (imprinting on their home signature), moving from general concept to specific mechanism across the three sentences.

14. Placing the explanation first gives the reader the background needed to understand why the study's results are significant; presenting the evidence first, without that context, would make the study's findings harder to interpret.

15. Paragraph 2 is structured to build understanding of a new concept step by step; paragraph 5 is structured to introduce doubt and then explain its implications, serving a different purpose — testing and qualifying the theory rather than simply explaining it.

Answer Key (4 of 4)

16. Removing paragraph 5 would make the article read as more one-sided and fully settled, losing the added credibility that comes from acknowledging open questions and showing the theory is still being tested and refined by researchers.

17. They had a name for the ability, "natal homing," but not a full explanation for how it worked.

18. Paragraph 1 ends by naming the unexplained phenomenon (natal homing) without a mechanism, and paragraph 2's topic sentence directly answers that gap by introducing magnetoreception as the likely explanation.

19. "Despite these open questions, magnetoreception remains the leading explanation..." is the topic sentence of paragraph 6; unlike the earlier topic sentences, which each introduce a new piece of the explanation or a complication, this one functions as a summary that reaffirms the overall theory after weighing its complications.

20. Paragraph 4 presents supporting evidence for the theory (the 2015 study's results), and paragraph 5 immediately follows by testing the theory's limits (the complication of magnetic drift), mirroring how real scientific reasoning builds support for an idea and then honestly examines where that idea might not fully hold.

Exit Ticket

Answer on a sticky note

Answer the exit-ticket question from today's lesson.

Cite at least one piece of evidence from the passage.

Remember

Name your claim or answer

Quote or cite the evidence

Explain how it supports your answer